

Software Requirements Specification (SRS)

AI Poem Visualizer — v2.0

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1. Introduction

1.1 Purpose

This SRS defines the functional and non-functional requirements for the AI Poem Visualizer — a web application that transforms user-submitted poems into AI-generated visual artwork.

1.2 Scope

The system accepts poems in English, Hindi, and Marathi; performs NLP analysis; generates matching background images via AI; composes typography-overlaid final artwork; and provides downloadable output in multiple formats.

1.3 Definitions

Term	Definition
NLP	Natural Language Processing
SD	Stable Diffusion (local image generation model)
JWT	JSON Web Token (stateless authentication)
Provider	An AI backend (Gemini, Pollinations, SD)

2. System Overview

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User → Frontend SPA → FastAPI Backend → AI Provider Layer → Output Images

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SQLite/PostgreSQL DB

(users, poems, generations, downloads, logs)

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3. Functional Requirements

3.1 Guest Users

- FR-G-01: Submit a poem (max 1200 characters) in English, Hindi, or Marathi
- FR-G-02: Select theme override (romantic, nature, dark, joyful, mystical, melancholic)
- FR-G-03: Select output format (1:1 square, 9:16 story)
- FR-G-04: Select image provider (Stable Diffusion local, Pollinations cloud)
- FR-G-05: Configure SD parameters (style, size, steps, guidance, seed)
- FR-G-06: View generated artwork in preview
- FR-G-07: Download artwork as PNG
- FR-G-08: View public gallery of recent creations

3.2 Registered Users

- FR-U-01: Register with name, email, password
- FR-U-02: Login with email, password; receive JWT
- FR-U-03: Logout (client-side token discard)
- FR-U-04: View and update display name
- FR-U-05: Save poems to personal collection
- FR-U-06: View saved poems, generate from saved, delete saved
- FR-U-07: View personal generation history with status and timing

3.3 Admin Users

- FR-A-01: View system-wide stats (users, poems, generations, downloads)
- FR-A-02: List all registered users with roles and status
- FR-A-03: View recent activity logs

3.4 AI Pipeline

- FR-P-01: Analyze poem → extract theme, mood, visual_prompt, language
- FR-P-02: Generate background image from visual_prompt
- FR-P-03: Compose final artwork (overlay poem text on background)

- FR-P-04: Support provider chain with automatic fallback
- FR-P-05: Log every AI provider call (timing, tokens, status)

4. Non-Functional Requirements

4.1 Performance

- NFR-01: Pollinations image generation < 15 seconds
- NFR-02: Gemini NLP analysis < 5 seconds
- NFR-03: Image composition < 3 seconds
- NFR-04: API response time < 200ms for non-AI endpoints

4.2 Security

- NFR-05: Passwords hashed with bcrypt (cost factor ≥ 12)
- NFR-06: JWT tokens signed with HS256
- NFR-07: SQL injection protection via SQLAlchemy ORM
- NFR-08: XSS protection via HTML escaping in frontend
- NFR-09: Rate limiting: 10 req/hr guests, 50 req/hr registered
- NFR-10: CORS restricted in production

4.3 Reliability

- NFR-11: AI provider chain with automatic fallback
- NFR-12: All database writes wrapped in transactions
- NFR-13: Graceful error handling with user-friendly messages

4.4 Scalability

- NFR-14: Database swappable from SQLite to PostgreSQL via DATABASE_URL
- NFR-15: AI provider abstraction allows adding new providers without code changes in routes
- NFR-16: Rate limiting store swappable from in-memory to Redis

5. Constraints

- Python 3.10+ required
- Stable Diffusion local mode requires ~4GB RAM
- No GPU required (CPU-only mode supported)
- Free tier: Gemini Flash API limits apply